POLARIS - Assistive Smart Cane with AI Chatbot for Navigation and Obstacle Detection

Association for Computing Machinery
Student Chapter

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OBJECTIVE:

- Provide real-time obstacle detection (10–20 cm) for visually impaired users.
- Deliver multi-modal feedback: active buzzer alerts and LCD distance display.
- Enable AI-powered conversation via Gemini 2.5 Flash chatbot.
- Give GPS turn-by-turn walking directions through the Google Directions API.
- Extend all features to a companion Android navigation app.

Flask ✦ Gemini Google Cloud

TECHNOLOGIES USED:

Microcontroller: ESP32 (Wi-Fi + Bluetooth)
Sensor: HC-SR04 ultrasonic
Display: LCD1602 I2C
AI Chatbot: Gemini 2.5 Flash API
Navigation: Google Directions API
Cloud: Google Cloud Console
Firmware IDE: Arduino IDE
Backend: Python IDLE
Mobile App: Android APK

CONCLUSION:

The Smart Cane achieves 96.8% obstacle detection accuracy and sub-1.8 s AI chatbot response time. At ~AED 113 in hardware cost, it is 15× cheaper than commercial alternatives. The integrated Gemini 2.5 Flash chatbot and GPS navigation app create a truly intelligent assistive device for daily independent mobility.

FEATURES:

- Real-time ultrasonic obstacle detection with three alert zones (CLEAR / CAUTION / DANGER).
- Active buzzer with distinct patterns per zone — faster beep = closer obstacle.
- LCD1602 shows live distance on Row 1 and navigation instructions on Row 2.
- Push-button activated Gemini 2.5 Flash AI chatbot — ask directions, weather, anything.
- Google Directions API provides walking turn-by-turn directions via voice and LCD.
- Android APK companion app with full chatbot + GPS navigation, voice in/out.

RESULTS:

Performance metrics — full system test:

Obstacle detection accuracy: 96.8% (150 trials)
Buzzer alert latency: < 60 ms
Directions API response: 0.9 s
Gemini API avg. response: 1.6 s (4G network)
GPS accuracy (Android phone): 3 – 8 m
APK voice recognition: 93.2%
Battery life (continuous): 6.1 hours
False positive rate: 3.2%
Hardware cost: ~AED 113 (~INR 2,540)

System comparison (vs commercial alternatives):

Feature	This Project	WeWALK	UltraCane
Obstacle detection	✓	✓	✓
AI chatbot	✓	✗	✗
GPS navigation	✓	Partial	✗
Android app	✓	Limited	✗
Cost (approx.)	AED 113	AED 1,800	AED 2,800

METHODOLOGY:

- HC-SR04 sensor polls every 50 ms
- ESP32 classifies zone (CLEAR / CAUTION / DANGER)
- Buzzer pattern + LCD Row 1 updates
- Button press → voice captured via mic
- Python backend sends to Gemini 2.5 Flash
- Response spoken via TTS + shown on LCD
- Android APK mirrors all functions + GPS

ASSISTANT APK

Assistant

CHATBOT INTERACTION:

User: How do I get to Dubai Mall?

Cane: Head north on Sheikh Zayed Rd 2.4 km, then right on Financial Centre Rd. Arrive in ~35 min.

User: Will it rain today?

Cane: No rain expected. Clear, 38 °C. Stay hydrated.

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